$$x_{H_2O} = \frac{1.67}{1.67 + 0.016} = \frac{1.67}{1.686} = 0.99$$

$$\therefore\ x_{H_2O} + x_{glucose} = 1$$

$$0.99 + x_{glucose} = 1$$

$$x_{glucose} = 1 - 0.99$$

$$= 0.01$$

3.10% $\frac{W}{V}$ means that 10g of solute in 100ml solution

$$\therefore \text{ amount of iodopovidone in } 1.5 \text{ ml} = \frac{10 \text{ g}}{100 \text{ ml}} \times 1.5 \text{ ml}$$

$$= 0.15 \text{ g}$$

4. $$\frac{\text{mass of dissolved solid}}{\text{mass of water}} \times 10^6$$

$$\frac{5 \times 10^{-3} \text{ g}}{1.05 \times 10^{3} \text{ g}} \times 10^6 = 4.76 \text{ ppm}$$

5. (a) mass of 1.5 moles of $CoCl_2$ = 1.5 × 129.9

= 194.85 g

194.85 g anhydrous cobalt chloride is dissolved in water and the solution is make up to one litre in a standard flask.

6. 6% $\frac{V}{V}$ aqeous solution contains 6mL of methanol in 100 ml solution.

∴ To prepare 500 ml of 6% $\frac{V}{V}$ solution of methanol 30mL methanol is taken in a 500 ml standard flask and required quantity of water is added to make up the solution to 500 ml.

6. $C_1V_1 = C_2V_2$

$$6 \text{ M } (V_1) = 0.25 \text{ M} \times 500 \text{ ml}$$

$$V_1 = \frac{0.25 \times 500}{6}$$

$$V_1 = 20.83 \text{ mL}$$

7. Total pressure = 1 atm

$$P_{N_2} = \left(\frac{80}{100}\right) \times \text{ total pressure } = \frac{80}{100} \times 1 \text{ atm} = 0.8 \text{ atm}$$

$$P_{O_2} = \left(\frac{20}{100}\right) \times 1 = 0.2 \text{ atm}$$

Accordingg to Henry's Law

$$P_{solute} = K_H\, x_{solute \text{ in solution}}$$

$$\therefore P_{N_2} = (K_H)_{Nitrogen} \times \text{mole fraction of } N_2 \text{ in solution}$$

$$\frac{0.8}{8.5 \times 10^4} = x_{N_2}$$

$$x_{N_2} = 9.4 \times 10^{-6}$$

Similarly,

$$x_{O_2} = \frac{0.2}{4.6 \times 10^4}$$

$$= 4.3 \times 10^{-6}$$

9. $P^o_{\text{pure benzene}} = 50.71$ mm Hg

$P^o_{\text{naphthalene}} = 32.06$ mm Hg

Number of moles of benzene = $\frac{39}{78}$ = 0.5 mol

Number of moles of napthalene = $\frac{128}{128}$ = 1 mol

mole fraction of benzene = $\frac{0.5}{1.5}$ = 0.33

mole fraction of napthalene = 1 – 0.33 = 0.67

Partial vapour pressure of benzene

= $P^o_{benzene}$ × mole fraction of benzene

= 50.71 × 0.33

= 16.73 mm Hg

Partial vapour pressure of napthalene

= 32.06 × 0.67

= 21.48 mm Hg

Mole fraction of benzene in vapour phase = $\frac{16.73}{16.73 + 21.48} = \frac{16.73}{38.21} = 0.44$

Mole fraction of napthalene in vapour phase = 1 – 0.44 = 0.56

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